

01 Project Overview and Objective

A concise view of the problem, scope, and proposed solution

AI Interview Pro

AI Interview Pro is an AI-assisted mock interview and candidate assessment platform for candidates, recruiters, and administrators. It combines resume analysis, structured interview sessions, multi-dimensional scoring, feedback, analytics, and consent-controlled reporting in one web application.

Project Objective

The objective is to help candidates practice and improve while giving authorized recruiters consistent evidence for review. Administrators receive the controls needed to monitor users, interview activity, AI behavior, and system health.

Existing System and Proposed System

Area	Typical Existing System	AI Interview Pro Provides
Interview process	Manual or static question sets with separate tools.	AI-generated mock interviews with domain and difficulty controls.
Assessment	Subjective review or a single overall mark.	Technical, communication, confidence, and professionalism scores.
Feedback	Delayed, general, or dependent on interviewer availability.	Question-level feedback, strengths, weak areas, and progress tracking.
Recruiter access	Broad access or disconnected reports.	Candidate-controlled sharing of selected interview evidence.
Monitoring	Limited session signals and fragmented reports.	Session alerts, behavior signals, dashboards, notifications, and reports.

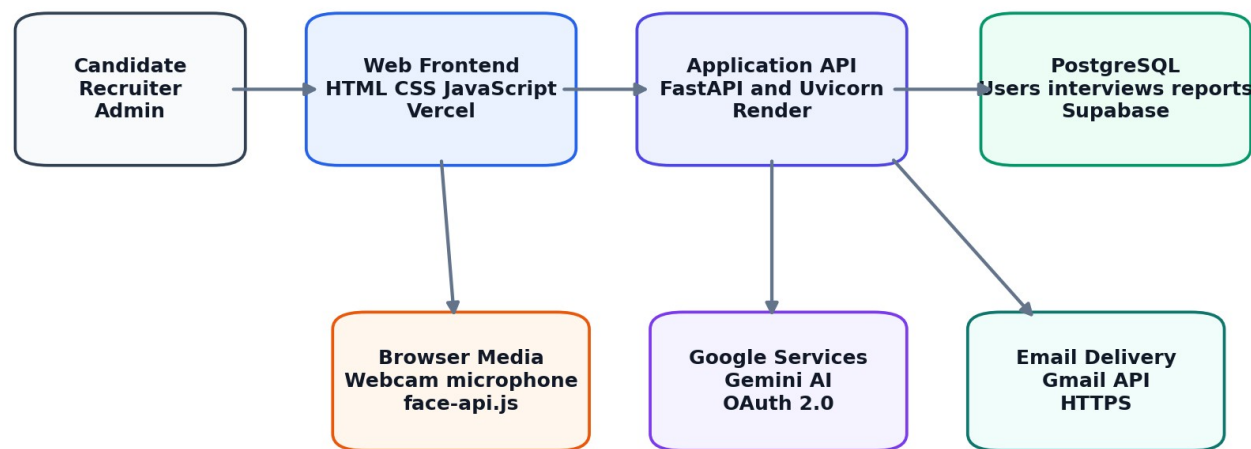
Project Scope

- Secure local and Google sign-in with role-based access
 - Resume upload, parsing, skill extraction, and interview setup
- Webcam and microphone interview sessions with rule monitoring
 - Scoring, AI feedback, analytics, sharing, reminders, and reports

02 System Architecture

Deployed components and the main data paths

The browser hosts the user interface and real-time media analysis. The FastAPI service applies authentication, business rules, scoring, consent checks, and integrations. PostgreSQL stores persistent application data.



GitHub stores the source and triggers frontend and backend deployments.

Figure 1 Deployed architecture and service integrations

Layer	Responsibility	Current Service
Presentation	Role dashboards, forms, charts, interview interface	Vercel
Application	FastAPI endpoints, validation, scoring, consent, reports	Render
Data	Users, resumes, interviews, scores, notifications	Supabase PostgreSQL
AI and identity	Question generation, feedback, Google sign-in	Gemini and Google OAuth
Messaging	Password-reset codes and enabled notifications	Gmail API

Data protection boundary Authenticated API requests protect dashboard data. Recruiter evidence endpoints also enforce candidate consent. Passwords for local accounts are hashed in the database; deployment secrets remain environment variables.

03 Candidate Dashboard and Flow

Practice, assessment, improvement, and controlled sharing

The candidate dashboard guides a user from resume preparation through interview practice and long-term improvement. Results come from completed interviews and uploaded resume data rather than fixed demonstration values.

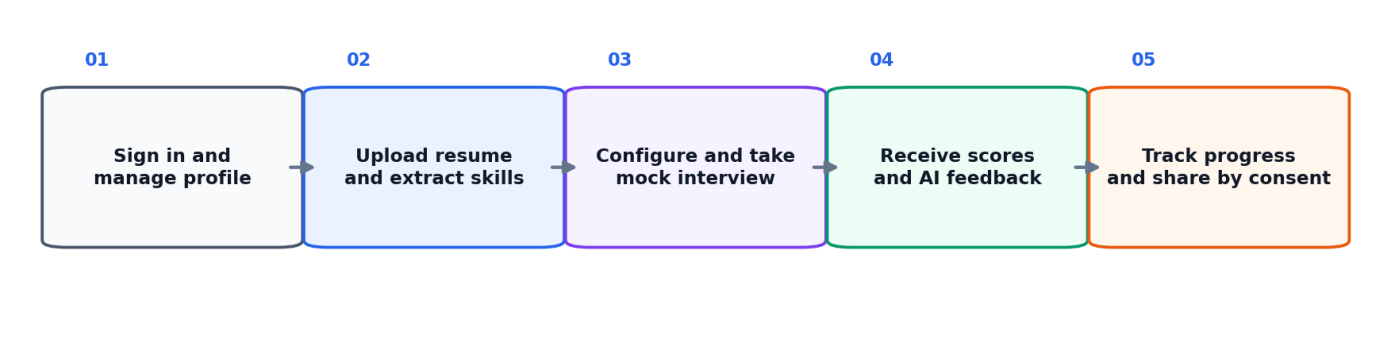


Figure 2 Candidate workflow

Dashboard Area	What the Candidate Can Do
Overview	View overall score, readiness, recent activity, and improvement progress.
Resume Analysis	Upload PDF or DOCX resumes and review extracted skills, education, experience, and summary.
Mock Interviews	Choose interview type, difficulty, domain, or resume skills; then use webcam and microphone.
Performance Analytics	Review score breakdowns, skill-wise results, weak areas, and performance trends.
History and Feedback	Open past attempts, question-level scores, AI feedback, and improvement guidance.
Reports and Sharing	Download reports and grant or revoke recruiter access to selected interview evidence.
Notifications and Profile	Manage reminders, session alerts, email preferences, account details, and saved logins.

Interview session behavior

The session requests explicit recording and assessment consent, captures answers, monitors fullscreen and face visibility rules, and saves provisional results quickly. AI feedback can complete in the background and update the report when available.

04 Recruiter Dashboard and Flow

Consent-scoped candidate review and hiring insights

The recruiter dashboard shows only candidates and interviews shared with the signed-in recruiter. It combines resume information, interview evidence, analytics, and comparison tools without exposing unshared candidate records.

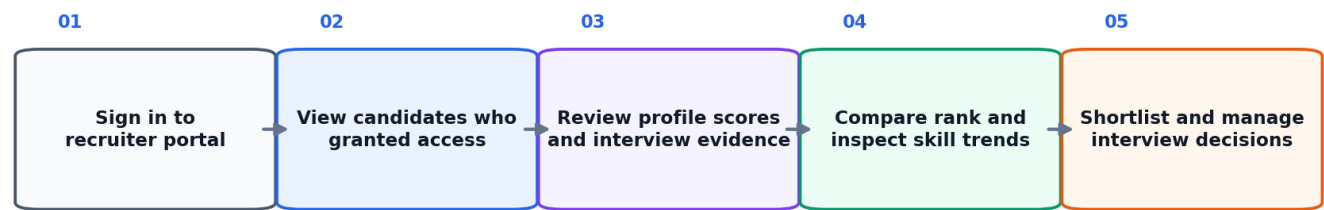


Figure 3 Recruiter workflow

Dashboard Area	Recruiter Function
Recruitment Overview	Monitor shared candidates, completed interviews, average score, and top score.
Candidate Profiles	Review permitted profile details, resume skills, score evidence, feedback, and recordings when available.
Comparison and Ranking	Compare consented candidates and rank them using shared interview performance.
Skill Analytics	Inspect technical, communication, confidence, and overall performance patterns.
Resume Screening	Review uploaded resume summaries and completeness indicators for visible candidates.
Interview Management	Review shared interviews and use evidence to support shortlisting decisions.
Reports and Notifications	Open performance reports, receive alerts, schedule reminders, and manage email preferences.

Decision support

Rankings and comparisons support recruiter review but do not replace human judgment. The dashboard presents the recorded evidence and makes the candidate's sharing status visible.

05 Admin Dashboard and Flow

Platform monitoring, management, and operational visibility

The admin dashboard provides platform-wide oversight. It brings together user management, interview monitoring, AI scoring indicators, feedback records, system health, and usage analytics in one administrative view.

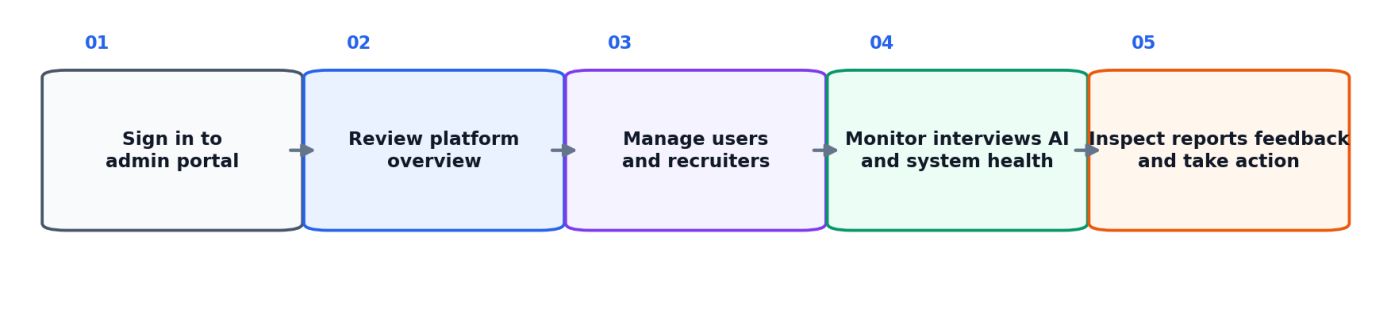


Figure 4 Administrator workflow

Dashboard Area	Administrator Function
Platform Overview	View user counts, role distribution, interview activity, and current operational indicators.
User Management	Review candidate, recruiter, and admin accounts and manage account status where supported.
Interview Monitoring	Inspect session activity, completion status, violations, and stored assessment records.
AI Performance	Review scoring dimensions, fallback behavior, feedback completion, and model-related activity.
System Health	Check database connectivity, media storage status, API readiness, and service activity.
Usage Analytics	Review platform adoption, interview volumes, trends, and role-level activity.
Reports and Feedback	Inspect administrative reports and user feedback, and manage alerts and notification preferences.

Administrative boundary

The admin role manages the platform rather than recruiter hiring decisions. Candidate-to-recruiter sharing remains controlled by the candidate consent workflow.

06 Technology Stack and End to End Summary

Implemented technologies, integrated workflow, and current limits

Layer	Technologies Used
Frontend	HTML5, CSS3, JavaScript, Canvas, Web Media APIs, face-api.js
Backend	Python, FastAPI, Uvicorn, SQLAlchemy, Pydantic
Database	PostgreSQL hosted on Supabase
AI and analysis	Google Gemini Flash, deterministic scoring rules, client-side facial signals, engagement model
Authentication	JWT sessions, Google OAuth 2.0, password hashing, password-reset verification
Notifications	In-app notifications, reminders, reports, Gmail API email delivery
Development and test	VS Code, Git, GitHub, Docker, Docker Compose, Postman, Pytest, GitHub Actions
Deployment	Vercel frontend, Render backend, Supabase PostgreSQL

Live deployment <https://smarthire-ai-interview-two.vercel.app>

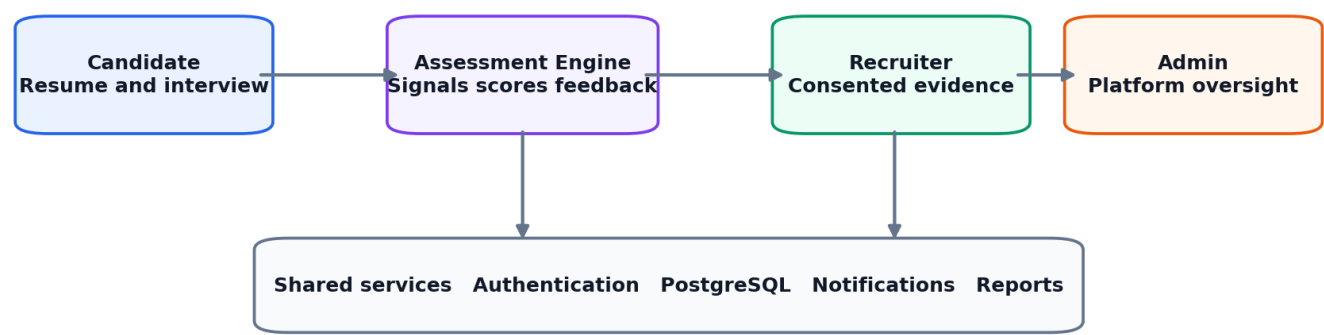


Figure 5 End-to-end role and evidence flow

Current Implementation Status

- Implemented: three role dashboards, resume analysis, interviews, scoring, analytics, consent sharing, notifications, reports, Google sign-in, password reset, and cloud deployment.
- Current limits: Render free-tier cold starts may delay the first request, and locally stored recordings do not persist across redeployments; cloud object storage is recommended.

Summary

The project delivers the complete practice-to-assessment workflow with role-based dashboards, consent-controlled recruiter access, AI-assisted feedback, reporting, and cloud deployment.

07 Model Performance Evaluation

Measured engagement-model accuracy, method, and validation limits

The saved engagement RNN was evaluated on 2,000 independently generated held-out synthetic behavior sequences: 500 each for engaged, distracted, absent, and multiple-person conditions.

Held-out synthetic behavior classification

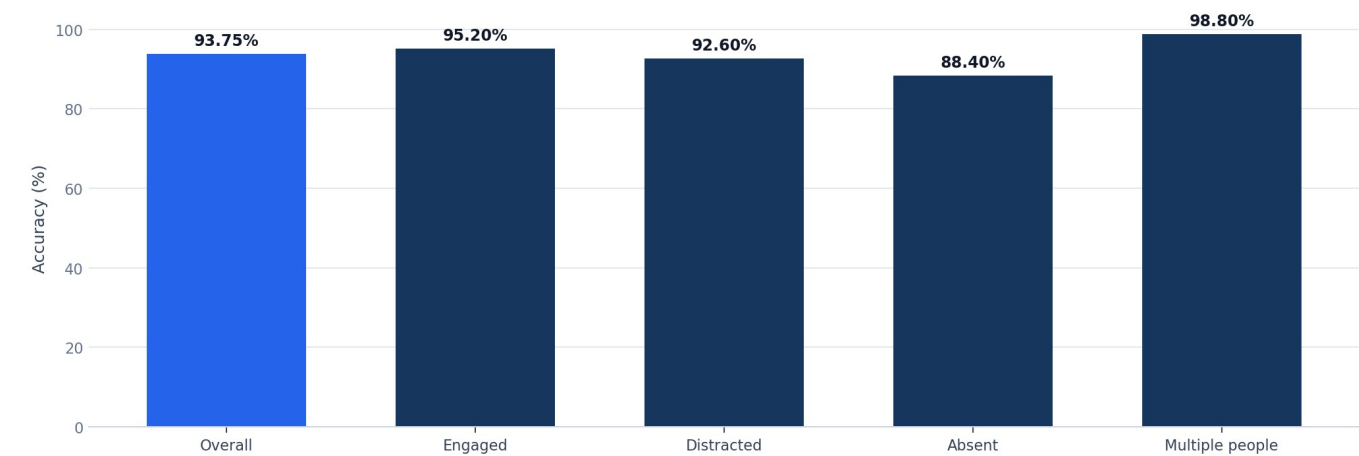


Figure 6 Engagement RNN classification accuracy by behavior class

Performance Metric	Result	Calculation Basis
Overall behavior classification accuracy	93.75%	1,875 correct classifications out of 2,000 test sequences
Normalized continuous-score agreement	90.97%	100 x (1 - overall mean absolute error)
Engagement output MAE	0.0993	Absolute difference between predicted and target engagement score
Disengagement-risk output MAE	0.1108	Absolute difference between predicted and target risk score
Integrity-risk output MAE	0.0609	Absolute difference between predicted and target integrity-risk score

Evaluation Method

- Load the trained engagement_rnn.npz weights and use a fixed evaluation seed for repeatability.
- Generate unseen sequences from the four documented behavior archetypes and predict three continuous outputs.
- Map each prediction to its closest behavior profile and calculate correct classifications divided by total sequences.

Interpretation and Limits

The 93.75% figure is the valid headline accuracy for the engagement RNN under its controlled synthetic test. It is not evidence of real-world speech-transcription, facial-emotion, eye-contact, or interview-scoring accuracy. Those claims require a separately labelled human test dataset. The next validation step is to collect consented reviewer-labelled sessions, preserve a true test split, and report accuracy, precision, recall, F1 score, and a confusion matrix.